

Neha R

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SUMMARY

Final-year Integrated M.Tech (AI/ML) student at VIT Bhopal. Built three end-to-end AI systems — an agentic RAG pipeline (LangGraph + Pinecone), an offline multi-agent loan assistant, and a multimodal OCR pipeline — deployed with FastAPI and Docker. AWS Solutions Architect certified. Actively seeking AI Engineering roles.

TECHNICAL SKILLS

AI Engineering: LangGraph (State Machines), Agentic RAG, LLMs, Multi-Agent Orchestration

Vector DBs & Retrieval: Pinecone, FAISS, Sliced Embeddings (512-dim), Semantic Search, Chunking Strategies

Backend & APIs: Python, SQL (PostgreSQL/MySQL), FastAPI (Async/Await), REST APIs

Performance & Cloud: AWS (Solutions Architect Associate), Docker

PROJECTS

Agentic AI eBook Assistant |

Feb 2026

LangGraph, Python, FastAPI, Pinecone, Docker

- Built a state-machine based RAG system using LangGraph and reduced out-of-context responses on a 50-query eval set from 30% to under 5% using relevance scoring.
- Engineered an asynchronous Ingestion Pipeline that processes PDF data into 512-dimensional vectors stored in a Pinecone index for high-speed semantic retrieval.
- Developed a Multi-Node Logic Flow (Retriever, Relevance Checker, Generator, Validator) to provide deterministic, traceable agent behavior and real-time Gradio interaction.
- Achieved <2s Response Times by leveraging FastAPI's non-blocking I/O operations and optimized vector similarity search.

Agentic Loan Workflow Automation |

Dec 2025

Python, FastAPI, Gemini, Docker

- Built a completely offline, multi-agent AI system that automates the entire loan application lifecycle, acting as a virtual bank worker from the first customer chat to printing the final approval letter.
- Engineered a team of specialized AI agents using Python to handle different tasks automatically, including sales pitching, identity verification, and financial risk calculation.
- Guaranteed 100% data privacy and reduced processing time by 40% by ensuring no sensitive documents leave the device; utilized Model Quantization to reduce memory footprint by 60%.

AutoForm AI — Multimodal Document Processing |

Nov 2025

Python, FastAPI, spaCy NLP, OpenCV, Docker

- Built an offline data engineering pipeline combining image preprocessing and transformer-based recognition, reducing processing latency by 78%.
- Eliminated manual data labelling by building an intelligent field classification system using spaCy NLP and regex patterns, automatically tagging extracted text into Name, Phone, Email, and ID categories with zero user input.
- Built a self-learning rule engine in the content script that silently maps form field labels to Vault keys on every manual input, persisting rules in Chrome local storage to skip fuzzy matching on future autofills.

EXPERIENCE

Salesforce Intern – Impaqtive

Oct 2025 – Dec 2025

- Gained hands-on exposure to Salesforce platform architecture — built Screen and Record-Triggered Automation Flows, configured custom objects and data schemas, and wrote SOQL queries for data retrieval across CRM modules.
- Explored Salesforce's AI layer (Einstein/Agentforce) and its agent-building paradigm — directly informed my approach to designing multi-agent AI systems with tool-use and state management in Python.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal

Aug 2022 – May 2027

Integrated M.Tech in Artificial Intelligence & Machine Learning

CGPA: 8.69 / 10.0

Relevant Coursework: Deep Learning, Computer Architecture, Generative AI, Data Structures

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate** | Verified Badge 2026
- Salesforce Developer with Agentblazer Champion** 2025
- Cloud Computing – NPTEL 2024
- Applied Machine Learning in Python – University of Michigan 2023